Reproducible 6-DOF Wrist-Roll Retrofit for OpenMANIPULATOR-X

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***Abstract*—Low-cost educational manipulators lack the full orientation control required for complex manipulation tasks due to their limited degrees of freedom. This paper presents a reproducible retrofit that upgrades the OpenMANIPULATOR-X from a 5-DOF to a 6-DOF system by integrating an additional wrist-roll joint, together with a unified ROS 2/MoveIt 2 software stack and an open hardware kit. A pose-based benchmarking framework compares the 5-DOF and 6-DOF configurations under identical conditions across kinematic feasibility, constrained-corridor planning, and physical hardware execution.**

Results show the 6-DOF arm achieves complete reachability across the tested orientation set (7/7) versus 1/7 for the 5-DOF baseline. In a constrained corridor at the narrowest tested width, the 6-DOF configuration achieves a 24.3% planning success rate versus 0.0% for the 5-DOF system ($\chi^2(1)$=17.14, p=3.5×10^{-5}, Cohen's h=1.031), with success rising above 80% at moderate corridor widths. Hardware validation on 120 poses, performed on a single retrofitted hardware prototype, shows the retrofit preserves orientation fidelity on physical hardware (97.5% success within 5°, mean error 1.89°) despite a larger, honestly quantified Cartesian position gap (40% strict success within 5 mm). The complete hardware and software design is released as an open, low-cost (≈$58 incremental) upgrade path intended to support replication by other laboratories; reproducibility claims in this paper refer to the design and software artifacts, while hardware results reflect validation on the single physical prototype described above.

***Index Terms*—Degrees of freedom, robot kinematics, OpenMANIPULATOR-X, wrist-roll retrofit, ROS 2, MoveIt 2, motion planning, benchmarking.**

I. Introduction

Low-cost educational manipulators such as the OpenMANIPULATOR-X [1] are widely used with ROS 2 [2] and MoveIt 2 [3] for instruction and prototyping, but their standard 5-DOF configuration lacks an independent wrist-roll joint, restricting the orientations achievable at a fixed end-effector position. This constrains tasks requiring precise alignment or orientation-controlled manipulation, where compensatory joint motions increase execution time and reduce planning success. Six independent DOF are theoretically required for arbitrary end-effector pose (position and orientation) [4], [5]; at a fixed position, achieving arbitrary orientation specifically requires an independently actuated wrist-roll axis. This requirement's practical impact on low-cost platforms is rarely characterized under controlled, reproducible conditions.

This paper presents a fully reproducible 6-DOF wrist-roll retrofit for the OpenMANIPULATOR-X: mechanical integration, an updated URDF/Xacro model, `ros2_control` and MoveIt 2 configuration, and a pose-based benchmarking framework comparing 5-DOF and 6-DOF configurations under identical conditions.

Contributions:

- A low-cost (≈$58 incremental) 6-DOF wrist-roll retrofit with open, reproducible hardware and software artifacts, validated in simulation and on a single physical prototype.
- A pose-based benchmarking framework that isolates the effect of the added DOF across IK feasibility, constrained planning, and hardware execution.
- Statistically grounded evidence (chi-square, effect sizes, confidence intervals) of the retrofit's benefit, together with an honest, quantified sim-to-real comparison.

Section II reviews related work; Section III describes the retrofit; Section IV details the methodology; Section V reports results; Section VI discusses implications; Section VII concludes.

II. Related Work

Manipulators spanning a range of cost and capability — from low-cost platforms such as OpenMANIPULATOR-X [1] and WidowX [6] to research-grade arms such as the Franka Panda [7] — are widely used in robotics education and research [8]; among the low-cost tier, 5-DOF platforms face inherent orientation-coverage limits, discussed further below. Six DOF are required for arbitrary end-effector pose [4], and the absence of an independent wrist-roll joint is known to prevent full orientation specification [4], [5]; this limitation is, however, typically discussed qualitatively rather than characterized experimentally on widely used low-cost hardware. Modern manipulation stacks build on ROS 2 [9], `ros2_control`, and MoveIt 2 [3], with OMPL [10] providing sampling-based planners, Planner Arena [11] supporting systematic benchmarking of planning algorithms, and standardized protocols such as those of Falco *et al.* [12] supporting systematic benchmarking of manipulator and end-effector performance more broadly. Constraint-aware planning in narrow or cluttered environments has been studied by Berenson *et al.* [13], highlighting the role of manipulator dexterity under geometric restriction. Reproducibility-focused efforts such as Meta-World [14] emphasize sharing complete experimental artifacts; however, prior low-cost-manipulator retrofits typically report hardware augmentation without a fully integrated, statistically validated evaluation pipeline. This work addresses

TABLE I
INCREMENTAL BILL OF MATERIALS (6-DOF RETROFIT)

Component	Qty	Notes
Dynamixel XL430-W250-T	1	Wrist-roll actuator, ID 15
Wrist link + gripper bracket	2	3D-printed adapters
Fasteners + RS-485 cable	—	Standard hardware
Total incremental cost: ≈5,599 INR (≈$58)		

that gap by combining a documented mechanical/software retrofit with a controlled, pose-based statistical comparison.

III. SYSTEM DESIGN

A. Mechanical Retrofit

A single revolute joint is added between the wrist-pitch link (*link5*) and the gripper assembly, actuated by a Dynamixel XL430-W250-T servo (ID 15; the existing gripper servo is reassigned to ID 16 to avoid conflicts). This achieves independent end-effector roll without modifying the five existing joints, at an incremental hardware cost of ≈$58 (Table I) and ≈1 hour of fully reversible assembly; both structural adapters are 3D-printable, and STL/STEP files are provided in the project repository (link withheld for double-blind review; released upon publication).

B. Software Integration

The URDF/Xacro kinematic chain was extended with a revolute joint (axis $[1, 0, 0]$, limits $\pm\pi$ rad) between *link5* and a new `wrist_roll_link`, with downstream gripper and TCP frames re-parented accordingly. The `ros2_control` hardware interface, transmission matrices, and controller configuration were expanded from five to six joints, and the MoveIt 2 SRDF, joint limits, and controller lists were updated to include `joint5_roll` in the arm planning group. A GUI panel and CSV logging were extended to expose the new joint. All modifications and per-commit diffs are available in the project repository for full reproducibility.

IV. METHODOLOGY

A. Pose-Based Evaluation and Workspace Estimation

Initial joint-space sampling was found to unfairly favor the 6-DOF arm's larger configuration space; evaluation was therefore switched to a pose-based framework in which both arms are compared on their ability to reach identical target end-effector poses. Each configuration's reachable workspace was estimated via forward-kinematics sampling (convex-hull and alpha-shape boundaries, Fig. 1), and the intersection of the two workspaces was used to constrain all subsequent pose generation, so that every evaluated pose is reachable by both arms independent of positional workspace differences. The alpha-shape parameter and forward-kinematics sampling density used to estimate each workspace are documented in the project repository to enable exact reproduction.

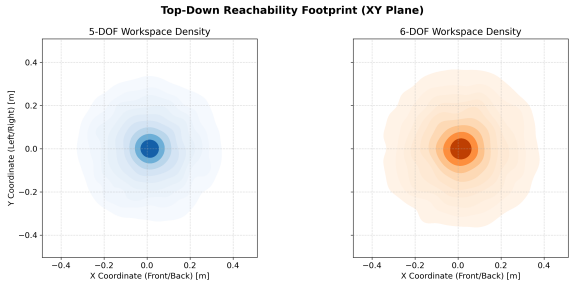


Fig. 1. Shared reachable workspace (top-down density) used to constrain all evaluation poses for both configurations.

B. Orientation Reachability Protocol

At a fixed, collision-free position $(0.20, 0.00, 0.15)$ m within the shared workspace, seven pure-roll orientations (0°–90°, 15° steps) were evaluated via the MoveIt 2 IK service, isolating IK feasibility from motion-planning effects. This discrete grid characterizes coverage trends across a single rotational axis rather than certifying continuous full-orientation feasibility; denser or higher-dimensional sampling is left to future work.

C. Constrained-Corridor Planning Benchmark

A narrow corridor scene was built in the MoveIt 2 planning scene, with width swept over five values (0.24–0.40 m) to characterize planning success as a function of spatial constraint (Fig. 2). At each width, goal poses were restricted to the IK-feasible set from IV-B, and planning was performed with OMPL RRTConnect (10 s limit, 10 trials per goal) to account for planner stochasticity, a sample size consistent with common practice in OMPL/MoveIt 2 benchmarking studies for characterizing binary success rates under repeated stochastic planning attempts. At the narrowest tested width, both configurations were evaluated over 7 feasible goals × 10 trials/goal (70 trials), giving a directly comparable sample size across the two configurations.

D. Cross-Configuration Kinematic Compatibility

With collision checking disabled, two independent sets of ∼8,000 end-effector poses were generated via forward-kinematics sampling of each arm's own joint space (Halton low-discrepancy sequence) and evaluated for IK solvability on *both* arms. IK solvability was evaluated with the same MoveIt 2 IK service and solver tolerance settings used elsewhere in this study; exact position and orientation tolerance values are fixed in the solver configuration file released in the project repository, allowing exact reproduction without restating them here. This large sample is used to densely characterize the shape of each arm's reachable-orientation manifold, and is distinct from the smaller, trial-based samples in Section V used for planning-success hypothesis testing. Because the 5-DOF arm's feasible roll at any fixed position is a single value, while poses in this test are drawn from a continuous $\pm\pi$ rad range, near-zero overlap between the two sampled manifolds is the expected outcome under a tight IK

TABLE II
IK FEASIBILITY BY ROLL ORIENTATION

Config.	0°	15°	30°	45°	60°	75°	90°	Total
5-DOF	✓	×	×	×	×	×	×	1/7
6-DOF	✓	✓	✓	✓	✓	✓	✓	7/7

tolerance almost by construction — analogous to asking how many points drawn uniformly from a cube land exactly on one face — rather than an inherently surprising empirical result.

E. Statistical Framework

Pearson's χ^2 test, Cramer's V, Cohen's h, and Wilson-score 95% confidence intervals were applied to binary planning-success outcomes; Welch's t-test/Cohen's d were used for planning-time comparisons. Cohen's h was computed via the arcsine-transform formula, $h = 2\arcsin(\sqrt{p_1}) - 2\arcsin(\sqrt{p_2})$, so that the reported value can be verified directly from the two reported proportions. Given a zero-count cell in the corridor-planning comparison (Table IV), Yates' continuity-corrected χ^2 was used; the uncorrected χ^2 statistic for the same table is higher ($\approx$19.35). All computations used Python (NumPy/SciPy).

F. Hardware Validation Protocol

A set of 120 validation poses (Halton sampling, then boundary/interior stratification via Farthest Point Sampling) was executed on both the simulated and physical retrofitted arm under identical `ros2_control` trajectory interfaces, recording joint tracking error, Cartesian position error, orientation error, and repeatability for direct sim-to-real comparison. All hardware results in Section V-E were collected on a single retrofitted physical unit; see the Limitations paragraph in Section VI for the scope this implies.

V. RESULTS

A. Orientation Reachability

The 5-DOF arm achieves a feasible IK solution only at 0° roll — a structural rather than algorithmic limitation, since no IK solution exists for non-zero roll at a fixed position without an independent wrist-roll joint. The 6-DOF arm achieves complete reachability across the tested orientation set, a 7× increase in coverage (14.3% → 100%), consistent with the theoretical requirement of six independent DOF for arbitrary end-effector pose [4], [5], of which the added roll axis specifically enables orientation coverage at a fixed position.

B. Cross-Configuration Kinematic Compatibility

Because the 5-DOF arm's feasible roll at any fixed position is a single value, while the 6-DOF-sampled poses are drawn from a continuous $\pm\pi$ rad range, near-zero overlap between the two manifolds is the expected outcome of this sampling design under a tight IK tolerance (Section IV-D) — not, on its own, a surprising empirical finding. Consistent with that expectation, the 5-DOF arm solves 94.13% of its own densely-sampled poses (confirming solver reliability) but 0% of the

TABLE III
CROSS-CONFIGURATION KINEMATIC COMPATIBILITY

Condition	Poses	Solved	Rate
5-DOF → 5-DOF (self)	7339	6908	94.13%
6-DOF → 5-DOF (cross)	7239	0	0.00%
Shared kinematic space			0.00%
Dexterity expansion (6-DOF vs. 5-DOF)			100.00%

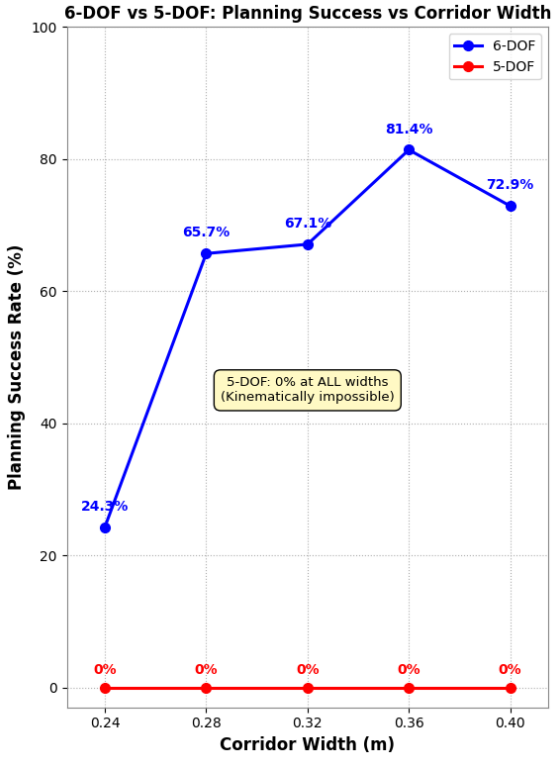


Fig. 2. 6-DOF vs. 5-DOF planning success across the full corridor-width sweep (0.24–0.40 m).

$\sim$7,239 poses drawn from the 6-DOF arm's manifold: no sampled 6-DOF pose was jointly IK-solvable by the 5-DOF arm under our solver tolerance. This corroborates, rather than independently establishes, the orientation-reachability result of Section V-A, and confirms the same effect holds under dense sampling of the full manifold rather than only at the seven discrete test orientations.

We summarize this as a *Shared Kinematic Space* of 0% (the fraction of the 6-DOF-sampled set also solvable by the 5-DOF arm) and a *Dexterity Expansion* of 100% (Table III). Given the sampling argument above, these bounds are expected rather than a novel discovery; we report them as a compact, quantified summary of the manifold comparison, relative to the sampled pose set and stated solver tolerance, not as a universal property of the two kinematic chains.

TABLE IV
CORRIDOR PLANNING RESULTS (WIDTH 0.24 M) AND STATISTICS

Config.	Trials	Successes	Rate (95% CI)
6-DOF	70	17	24.3% (15.8–35.5%)
5-DOF	70	0	0.0% (0.0–5.2%)

$\chi^2(1)$=17.14 (Yates-corrected), p=3.5×10^{-5}; Cramer's V=0.350; Cohen's h=1.031.

C. Constrained-Corridor Planning

At the narrowest tested width (0.24 m), the 0°-roll goal — the only orientation kinematically reachable by the 5-DOF arm — yields 0% planning success for *both* arms, indicating that the 6-DOF advantage at this width is entirely attributable to planning success at non-zero-roll goals, which are exclusively reachable by the 6-DOF configuration. The 24.3% headline figure therefore reflects a deliberately conservative (narrowest) test condition rather than the platform's best-case planning performance.

As shown in Fig. 2, success rises with corridor width for both configurations, peaking at 81.4% for the 6-DOF arm at 0.36 m before a slight decline at the widest tested setting (0.40 m, 72.9%); a plausible mechanistic explanation for this non-monotonic trend is discussed in Section V-D, though it was not directly measured in this study. The chi-square test rejects independence between configuration and planning success (p=3.5×10^{-5}), and Cohen's h=1.031 places the effect in the large category.

D. Joint Utilization

Analysis of successful 6-DOF trajectories (Fig. 3) shows `joint5_roll` values spread broadly across the ±π rad range, with no successful trajectory holding the joint near zero throughout execution — confirming the planner genuinely exploits the added DOF rather than leaving it kinematically passive. Successful trials are concentrated at the higher-roll orientations (60°–90°); at low roll (0°–45°), planning success is negligible regardless of corridor width. We hypothesize that this pattern reflects corridor geometry: at high roll, `joint5_roll` likely turns the gripper approximately parallel to the corridor walls, reducing its effective distal cross-section and opening collision-free approach trajectories that would otherwise remain geometrically blocked when the gripper is held perpendicular to the walls. Widening the corridor would then further relax this constraint, which may explain why planning success continues to rise with corridor width even though the underlying benefit originates from orientation, not clearance, freedom. This explanation is consistent with the observed roll-success correlation, but we did not directly measure collision geometry, approach angle, or roll-conditioned clearance statistics in this study; confirming it quantitatively is left to future work.

E. Hardware Validation

Simulation achieves near-numerical-precision position accuracy, while the physical arm's mean Cartesian error is 7.85 mm

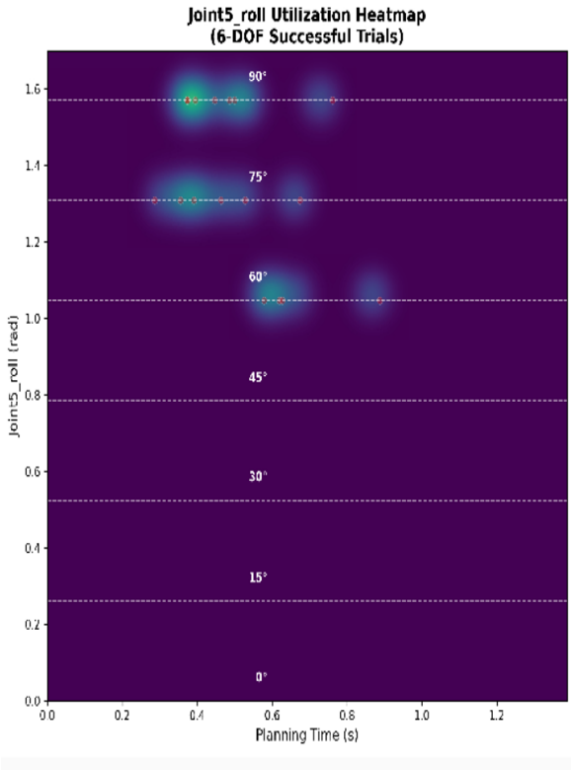


Fig. 3. Distribution of `joint5_roll` values across successful 6-DOF trajectories, confirming active use of the added joint.

TABLE V
HARDWARE VALIDATION: SIMULATION VS. REAL ROBOT (120 POSES)

Metric	Sim	Real	Gap
Strict position success (<5 mm)	100.0%	40.0%	−60.0 pp
Orientation success (<5°)	100.0%	97.5%	−2.5 pp
Mean Cartesian error	0.050 mm	7.849 mm	×156
Boundary success rate	100.0%	19.44%	−80.6 pp
Interior success rate	100.0%	48.81%	−51.2 pp
Repeatability (σ)	±0.0055 mm	±1.032 mm	×188
Mean angular error	0.014°	1.888°	×135

(40.0% strict success within 5 mm), attributable to Dynamixel gear backlash, structural compliance, and gravity loading — factors absent from the idealized model. Boundary poses, which impose higher joint torque, show a lower success rate (19.44%) than interior poses (48.81%), consistent with this interpretation. Despite the position gap, orientation fidelity is well preserved on hardware: 97.5% of executions satisfy the 5° constraint, with a mean angular error of only 1.89°. Repeatability (±1.03 mm) falls within the range reported for comparable Dynamixel-based platforms. All figures in this subsection reflect a single physical retrofit unit (Section IV-F); cross-unit variance is not characterized here.

VI. DISCUSSION

The IK-feasibility, cross-configuration, and corridor-planning results provide consistent evidence for the same

underlying structural effect: removing wrist roll eliminates feasibility for non-zero-roll goals. The corridor result is the one that translates this into a task-level planning benefit rather than a purely kinematic one, and — as Fig. 2 shows — this benefit strengthens substantially as spatial constraint is relaxed, a trend consistent with the hypothesized high-roll-orientation mechanism discussed in Section V-D, though that mechanism itself was not directly measured.

The hardware results distinguish two failure modes with different practical weight. Position accuracy on the physical arm is meaningfully worse than in simulation, largely due to actuator-class factors (Dynamixel backlash, compliance) rather than the retrofit itself. Orientation accuracy — the specific capability this retrofit adds — degrades far less, since `joint5_roll` is a compact, distal joint that carries less load than the proximal shoulder and elbow joints. For applications where the primary benefit sought is orientation freedom, the physical arm delivers it at an operationally useful level.

Practical implications. Within the evaluated orientation-reachability and corridor-planning benchmarks, the additional DOF determines whether the tested goals are reachable at all, rather than merely improving performance on goals reachable by both configurations; extending this to concrete manipulation tasks (e.g., insertion, cap alignment) is left to future work. The retrofit is low-cost ($\approx\$58$, $\approx$1 hour) and fully documented in open artifacts, supporting replication.

Limitations. Hardware validation was performed on a single physical unit; inter-unit variance from manufacturing tolerance and per-unit calibration is not characterized, so the reproducibility claim in this paper applies to the design and software, not yet to cross-unit hardware performance. Statistical treatment of the hardware results (Table V) is limited to point estimates and standard deviations rather than the confidence-interval framework used for the planning results; extending it is straightforward future work. The Shared Kinematic Space and Dexterity Expansion figures in Section V-B are reported relative to the sampled pose set and stated IK tolerance rather than as universal properties of the two kinematic chains. The corridor-planning mechanism proposed in Section V-D is a hypothesis consistent with the observed data, not a directly measured collision-geometry result. Finally, this study benchmarks the retrofit only against its own 5-DOF baseline; a direct comparison with existing commercial 6-DOF educational manipulators is left to future work.

VII. Conclusion

This work presented a low-cost, reproducible 6-DOF wrist-roll retrofit for the OpenMANIPULATOR-X and evaluated it through kinematic analysis, statistically grounded planning benchmarks, and hardware validation. The retrofit expands orientation coverage from 14.3% to 100% and improves constrained-corridor planning success from 0.0% to 24.3–81.4% depending on spatial constraint, while preserving orientation fidelity on physical hardware. Future work will evaluate roll-critical manipulation tasks (insertion, alignment), extend the statistical framework to hardware validation, and assess a second physical build for cross-unit reproducibility.

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